SMARTCAMPUS LOGIN & ACCESS CONTROL MODULE

Technical Specification & Implementation Guide — v1.0 · March 2026
Confidential · Internal Documentation

Audience: Development Team · Security Team · System Administrators
Related Modules: System Configuration · Student Management · Finance · Academics · HR · Communication

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1. MODULE OVERVIEW

The **Login & Access Control Module** is the **gateway to the SmartCampus platform**. It handles authentication, role-based access, personalization, and session management.

Objectives:

- Securely authenticate users
- Route users automatically based on role and permissions
- Filter modules, menus, and data based on access rights
- Maintain audit logs for all login and access events
- Support multi-role, multi-tenant, and multi-campus scenarios

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2. CURRENT GAPS

- Generic login requires manual role selection or module navigation.
- Users see menus/modules they cannot access (risk of confusion/exposure).
- Limited personalization; no session continuity.
- Minimal security for sensitive roles; no MFA or audit tracking.

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3. FUNCTIONAL REQUIREMENTS

3.1 Authentication

- **Inputs:** School ID, Username, Password
- **Outputs:** Auth token/session with associated roles and permissions
- **Enhancements:**
    - Multi-factor authentication (SMS/Email/Authenticator app) for high-sensitivity roles
    - Integration with Single Sign-On (SSO) if the school uses external identity providers
    - Audit logging of all login attempts (successful and failed)

3.2 Auto Role Login

- After authentication, the system evaluates **user roles**:
    1. **Single-role user** → Direct to assigned module
    2. **Multi-role user** → Dashboard shows only permitted modules
    3. **No assigned role** → Login succeeds; empty dashboard (security by obscurity)

3.3 Role-Based Dashboard Filtering

- Display **only modules the user is authorized for**
- Hide non-permitted modules completely (not greyed out)
- Permissions apply at **module level** and **data level** (RBAC/ABAC)

3.4 Session Personalization

- Track last accessed module per user → auto-redirect on next login
- Allow **quick role switch** for multi-role users
- Persist user preferences (theme, notifications)

3.5 Security

- Implement **audit logging**: track user, role, module accessed, timestamp
- Optional **multi-factor authentication** for sensitive roles (Finance, Admin)
- Protect against session hijacking and replay attacks
- Enforce **strong password policies** and optional SSO integration

3.6 Scalability & Integration

- API-based role assignment for integration with HR or Student Information Systems
- Multi-campus support with **tenant-aware roles**
- Support SSO across modules and external systems

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4. SYSTEM ARCHITECTURE

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[User Login Page]
      |
      v
[Authentication Service] --> Verify Credentials + MFA
      |
      v
[Role Resolver Service] --> Fetch RBAC/ABAC Permissions
      |
      v
[Module Router]
      |          \
      v           v
[Single Module] [Dashboard (multi-role)]
      |
      v
[Access Filter Service] --> Only allowed modules and data visible
      |
      v
[Session Store] --> Persist last accessed module and preferences
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5. DATA MODEL

5.1 User Table

Field	Type	Description
user_id	UUID	Unique system ID
school_id	UUID	Tenant/School reference
username	String	Login name
password_hash	String	Hashed password
mfa_enabled	Boolean	Flag for MFA
last_login	Timestamp	Last successful login
last_module	String	Last accessed module

5.2 Role Table

Field	Type	Description
role_id	UUID	Unique role identifier
role_name	String	Teacher, Parent, Finance, Admin, etc.
permissions	JSON	Modules + data-level access

5.3 Session Table

Field	Type	Description
session_id	UUID	Unique session token
user_id	UUID	User reference

active_roles	JSON	Roles active in session
last_accessed	Timestamp	Last module accessed

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6. SECURITY & COMPLIANCE

- Encrypted passwords (e.g., bcrypt/argon2)
- HTTPS for all communications
- MFA for sensitive roles
- Audit logs for compliance and investigations
- Session timeout and idle logout policies
- RBAC/ABAC enforcement to protect sensitive data

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7. USER EXPERIENCE FLOW

User Type	Login Flow
Parent	Auth → Auto-route → Parent Portal
Teacher	Auth → Auto-route → Classes/Reports
Teacher + Parent	Auth → Dashboard → Only permitted modules visible
Accountant/Finance Staff	Auth → Auto-route → Finance Module
Unassigned/No role	Auth → Dashboard empty → No modules displayed

UX Notes:

- Clean, minimal login page
- Role-specific branding (optional)
- Error handling: invalid login, inactive account, MFA failures

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